MILITARY SPECIFICATION
IGNITER

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers one type of igniter used to ignite a solid propellant.

2. APPLICABLE DOCUMENTS

2.1 Applicable documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the ext ent specified herein.

SPECIFICATIONS

Military

MIL-A-9836	Ammmunition Data Cards

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-S-633	Steel, Bars, Carbon, Cold Finished and Hot Rolled, (General Purpose)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-350	Jolt Test for Use in Production of Fuzes
MIL-STD-351	Jumble Test for Use in Production of Fuzes
MIL-STD-352	Forty (40) Foot Drop Test for Use in Production of Fuzes
MIL-STD-353	Transportation Vibration Test for Use in Production of Fuzes
MS-3192	Contact-Electrical, Pin, Crimp Removable 125°0
MS-33586	Metals, Definitions of Dissimilar

DRAWINGS

Bureau of Naval Weapons

DL-1330411	Container, Shipping and Storage (For Eight Igniters, DL 2526678)
2526678	Igniter
2526718	Packing, Preformed

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procuring functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent invoked herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal snail apply.

Code of Federal Regulations

49CFR 71-78	Transportation, Interstate Commerce Commission, Explosive and Other Dangerous Articles

(Application for copies shall be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders should cite the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the igniter are classified as follows:

REQUIREMENT	PARAGRAPH
General Requirements	3.2
Performance Requirements and Product Characteristics	3.3
Environmental Requirements	3.4
Operational Service Life	3.5

3.2 General requirements.

3.2.1 Data requirements. No data is required by this specification or approval documents referenced in Section 2 unless specified in contract or purchase order. See 6.2.

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3.2.2 Preproduction sample. Unless otherwise specified in the contract or purchase order, the preproduction sample of the igniter shall be manufactured using the methods and procedures proposed for production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining that prior to starting production. the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2.3 Materials.

3.2.3.1 Materials and treatments of materials. Unless otherwise specified in the contract or purchase order, all materials and treatments of materials and parts used in the manufacture of the igniter shall be in accordance with the applicable documents listed on BuWeps Dwg . 2526678.

3.2.3.2 Metals. Metals shall be of the corrosion-resistant type or shall be adequately protected to resist corrosion during the normal storage and service life of the igniter. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MS33586 shall not be used in intimate contact.

3.2.4 Definitions. Definitions of terms appear in Section 6.

3.3 Performance requirements and product characteristics. Unless specified otherwise, the igniter shall satisfy the following performance requirements when tested as specified in Section 4.

3.3.1 Static pressure. The igniter case, with the electrical terminals and insulating insert installed, shall be capable of withstanding an internal hydrostatic pressure of 10,000 pounds per square inch, gage (psig) for 10 seconds before and after firing, without leaking,

3.3.2 Bridge wire resistance. The igniter shall contain two resistive bridge wire elements, either one capable of firing the igniter. The resistance of each bridge element in a loaded igniter shall be 1.2 ± 0.15 ohms prior to firing. After firing, the resistance between pins 1 and 2 and between pins 3 and 2 shall be a minimum of 1.05 ohms at 1.5 volts, direct current.

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3.3.3 Insulation resistance. The resistance prior to firing, between the bridge wire and the case of the loaded igniter, snail be 1000 ohms minimum at 1.5 volts, direct current. The resistance between the bridge wire elements and the case after firing shall be 1.05 ohms minimum at 1.5 volts, direct current.

3.3.4 Auto-ignition.

3.3.4.1 Low temperature. The igniter shall be capable of withstanding a temperature of +170° F for a period of 2 hours without self-igniting and thereafter shall meet the performance requirements and product characteristics.

3.3.4.2 High temperature. The igniter shall be capable of withstanding a temperature of +250° F for a period of 8 hours without self-igniting. The igniter is not required, thereafter, to meet the performance requirements and product characteristics.

3.3.5 Maximum no-tire current. A direct current of 0.2 amperes maximum flowing through either bridge wire for a minimum of 5 minutes, shall not cause the igniter to fire, and thereafter shall meet the performance requirements and product characteristics.

3.3.6 Minimum firing current. Either bridge wire shall be capable of firing the igniter when a direct current of 0.9 amperes minimum is passed through it. The direct current shall be limited to an on-time of 100 milliseconds (ins).

3.3.7 Maximum firing current. Either bridge wire shall be capable of firing the igniter when a direct current of 5 amperes maximum is passed through it for 20 ms.

3.3.8 Pressure output. The igniter shall develop a pressure output versus time characteristic within the limits Illustrated in Figure 1 when fired into a closed chamber of 0.305 cubic inches with the igniter and chamber previously conditioned to a temperature in the range of 0° F to +140° F.

3.3.9 Calorimetric output. The total calculated calorimetric output shall be not less than 900 calories

3.3.10 Metallic particles. The metallic particle content shall constitute a minimum of 25 percent, by weight. of the main charge,

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3.3.11 Configuration.

3.3.11.1 Dimensions. Dimensions of the igniter shall be in accordance with BuWeps Dwg. 2526678 and all documents listed thereon.

3.4 Environmental requirements. Unless otherwise specified, the igniter shall be capable of conforming to the performance requirements and product characteristics of the specification after being exposed to the environmental conditions specified in this section. For definitions of ready, operating, and non-operating see 6.3.1, 6.3.2, 6.3.3.

3.4.1 Vibration, ready condition. The igniter, mounted as shown in Figure 2, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the vibrations defined in Table I. The igniter shall be vibrated for 180 minutes each in its longitudinal axis and in two mutually perpendicular transverse axes at any temperature within the range of -45° F to +140° F. Vibration is defined in 6.3.4.

TABLE I
VIBRATION SCHEDULE

Double Amplitude (in. ± 10%) Minimum	Acceleration (g's ± 10%) Minimum	Frequency (cps)
0.2	---	5 - 22
---	3.0	22 - 40
0.036	---	40 - 74
---	10	74 - 500

3.4.2 Acceleration. ready condition. The igniter, mounted as shown in Figure 2, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the accelerations defined in Table 11 for a duration of 1 minute for each magnitude. Acceleration is defined in 6.3.5.

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TABLE II

ACCELERATION SCHEDULE

Magnitude (g's ± 10%)	Acceleration Direction (See Figure 2)
25	Longitudinal axis, Fwd
35	Longitudinal axis, Aft

3.4.3 Shock, ready condition. The igniter, mounted as shown in Figure 2, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the shock loads defined in Table III. Shock is defined in 6.3.6.

TABLE III

SHOCK SCHEDULE

Test Sequence	Peak Magnitude (g's ± 10%)	Duration (ms)	Impact Axis	Direction Of Deceleration Force	Number Of Impacts
1	300	0.5 to 1.5	Longitudinal	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	18.0 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

3.4.4 Temperature and humidity.

3.4.4.1 Non-operating.

a. Temperature. The igniter shall be capable of meeting the performance requirements subsequent to withstanding, in the non-operating condition, environmental temperatures between -45° F and +140° F. Temperature is defined in 6.3.7.

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b. Humidity. The igniter shall be capable of meeting the performance requirements subsequent to withstanding, in the non-operating condition, an environment of at least 95 percent relative humidity at +110° F for a minimum period of 8 hours. Humidity is defined in 6.3.8.

3.4.4.2 Ready, operating. The igniter, immediately after being conditioned for 1 hour at any temperature between 0° F and +140° F shall be capable of meeting the performance requirements at any environmental temperature between 0° F and +140° F.

3.4.5 Jolt. The igniter, in the non-operating condition, shall be capable of withstanding the jolt test requirements of MIL-STD-350. It is not required that the igniter be operable after testing. The criteria, by which the samples are judged to have withstood the testing, are that (1) no explosive elements shall explode and (2) no parts shall be broken, be deformed, be displaced, come apart, or arm, in such a manner as to make the assembly unsafe to handle or dangerous to use as determined by examination.

3.4.6 Jumble. The igniter, in the non-operating condition, shall be capable of withstanding the jumble test requirements of MIL-STD-351. It is not required chat the igniter be operable after testing. The criteria, by which the samples are judged to have withstood the testing, are that (1) no explosive elements shall explode and (2) no parts shall be broken, be deformed, be displaced, come apart, or arm, in such a manner as to make the assembly unsafe to handle or dangerous to use as determined by examination.

3.4.7 Drop. The igniter, in the non-operating condition, shall be capable of withstanding the drop test requirements of MIL-STD-352. It is not required that the igniter be operable after testing. The criteria, by which the samples are judged to have withstood the testing, are that (1) no explosive elements shall explode and (2) no parts shall be broken, be deformed, be displaced, come apart, or arm, in such a manner as to make the assembly unsafe to handle or dangerous to use as determined by examination.

3.4.8 Transportation vibration. The igniter, in the non-operating condition, shall be capable of withstanding the transportation vibration test requirements of MIL-STD-353.

3.4.9 Accelerated storage life.

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3.4.9.1 Packed and packaged condition. The igniter, in the packed and packaged condition, shall be capable of withstanding an environmental temperature of +140° F for a period of 30 days. Thereafter, the igniter shall be capable of operating in accordance with 3.3.5. This requirement satisfies a five year storage life, in the packed and packaged condition.

3.4.9.2 Ready condition. The igniter, in the ready condition, shall be capable of withstanding an environmental temperature of +140° F for a period of 14 days. Thereafter, the igniter shall be capable of operating in accordance with 3.3.6. This requirement satisfies a one year storage life, in the ready condition.

3.5 Operational service life. The igniter shall have a service life of one duty cycle.

3.6 Ammunition data cards. Ammunition data cards shall be supplied in accordance with MIL-A-9836 and 6.2 e.

3,7 Workmanship . The workmanship and finish shall be of sufficiently high grade to insure satisfactory operation consistent with the requirements of this specification. The assembly shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribes requirements.

4.2 Classification of inspection. Inspection of the igniter shall be classified as follows:

a. Preproduction inspection (Subsection 4.4)
b. Acceptance inspection (Subsection 4.5)
c. Periodic inspection (Subsection 4.6)

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4.3 Sampling.

4.3.1 Preproduction inspection sample. A preproduction sample of 40 igniters manufactured in accordance with 3.2 shall be subjected to the preproduction tests listed in Table IV and detailed in 4.4 through 4.4.4.20 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. See 4.4.6. Further production of the igniter by the supplier, prior to approval of the preproduction sample shall be at the supplier's risk.

4.3.2 Acceptance Inspection sample. The sampling plans and procedures set forth in MIL-STD-105 shall be used in the determination of acceptability of the igniter, inspected as specified in 4.5 through 4.5.8 with the following exception: Each igniter shall be subjected to the tests detailed in 4.5.6.1 a., b., c., and d.

4.3.3 Periodic production sample. Periodic production samples shall be selected from units that have satisfactorily passed the acceptance inspections of 4.5.6.1. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production randomly select 9 units from each group of 150 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2.

Step 2. Randomly select 9 units from each group of 350 units. Continue with Step 2 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

d. Submit to the procuring activity for approval the proposed retest program to verify that the corrective action eliminated the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

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4.4 Preproduction inspection.

4.4.1 Tests and examinations. The preproduction sample of 40 igniters shall be inspected in accordance with Table IV, with 4 igniters being tested in accordance with each of the 10 test sequences, A through J.

TABLE IV

PREPRODUCTION TEST SEQUENCE

Tests	Paragraph Reference	Test Sequence A	B	C	D	E	F	G	H	I	J
Inspection, Prior to Firing	4.4.4.1	1	1	1	1	1	1	1	1	1	
Vibration, Ready Condition	4.4.4.2	2									
Acceleration, Ready Condition	4.4.4.3		2								
Shock, Ready Condition	4.4.4.4		3								
Temperature & Humidity	4.4.4.5			2	2						
Low Temperature Operating	4.4.4.6			3							
High Temperature Operating	4.4.4.7				3						
Auto-Ignition, Low Temperature	4.4.4.8					2	2	2			
Auto-Ignition, High Temperature	4.4.4.9									2	
Vibration, Packed and Packaged	4.4.4.10					3					
Shock, Packed and Packaged	4.4.4.11					4					
Jolt	4.4.4.12						3				
Jumble	4.4.4.13						4				
Drop	4.4.4.14						5				
Transportation Vibration	4.4.4.15							3			
Accelerated Storage Life-Packed and Packaged	4.4.4.16								2		
Accelerated Storage Life-Ready	4.4.4.17									2	
Minimum Fire Current	4.4.4.18		4			5			3	3	
Maximum No-Fire Current	4.4.4.19	3									
Maximum Fire Current	4.4.4.20	4						4			
Inspection, Post-Firing	4.4.4.1	5	5	4	4	6		5	4	4	

4.4.2 Test conditions. Unless otherwise specified, the igniter shall be subjected to the tests in 4.4.1 under the following conditions:

a. Temperature: Room ambient +32° F to +98° F
b. Altitude: Normal ground atmospheric pressure
c. Vibration: None
d. Humidity; Room ambient

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4.4.3 Test equipment and set-up. In addition to the items listed in 4.5.5, test equipment is required which provides the following characteristics:

4.4.3.1 Vibration equipment with temperature chamber. The vibration equipment with temperature chamber shall have the following characteristics:

a. Frequency range, logarithmic sweep: 5 to 500 cycles per second (cps)
b. Test specimen: 8 ounces
c. Acceleration and displacement (ready condition):

5 to 22 cps	0.2 inch
22 to 40 Cps	3.0 gravitational units (g's)
40 to 74 cps	0.036 inch
74 to 500 cps	10 g's

d. Acceleration (packed and packaged):

5 to 60 cps	1 g

e. Automatic cycling time range: 0 to 15 minutes (optional)
f. Temperature chamber range: -45° F to +140° F accuracy to ± 5°F
g. Interior size: 2 inches x 2 inches x 4 inches minimum

4.4.3.2 Acceleration equipment. The acceleration equipment shall have the following characteristics: 35 g's with an 8 ounce test specimen.

4.4.3.3 Shock equipment. The shock equipment shall be capable of accepting an 8 ounce test specimen and shall provide ranges of:

a. 300 g's for 0.5 milliseconds (ins) to 1.5 ms
b. 60 g's for 8.0 ms to 10.0 ms
c. 45 g's for 18.0 ms to 20.0 ms

4.4.3.4 Temperature and humidity conditioning equipment. One or more equipments may be utilized to provide the following characteristics:

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a.	Temperature:	-45° F to +250° F; with an accuracy to ± 5° F
b.	Relative humidity:	95 percent minimum at +110° F; with an accuracy to ± 5° F
c.	Interior size:	2 inches x 2 inches x 4 inches minimum

This equipment may be used for both non-operating and rearly/operating temperature (and humidity) tests.

4.4.3.5 Power supply. The power supply, in addition to meeting the requirements of 4.5.5 a., shall be capable of supplying a 100 ± 1 ms pulse.

4.4.4 Test procedures.

4.4.4.1 Inspection. The igniter shall be subjected to the tests, prior to firing , detailed in 4.5.6.1, and post-firing, detailed in 4.5.6.3.

4.4.4.2 Vibration, ready condition. The igniter, in the ready condition, shall be mounted in a manner equivalent to that shown in Figure 2. The resonant modes in each of the 3 axes shall be determined by varying the frequency of applied vibration slowly from 5 cps to 500 cps. The igniter shall then be subjected to resonance and cycling vibration in each axis. in conformance with the times shown in Table V. If more than 4 resonances are encountered on any axis, the 4 most severe resonances shall be chosen for the test. The resonant and cycling period, for each axis, shall be divided into 3 equal parts, which are conducted at -45° F ± 5° F, room temperature and +140° F ± 5° F respectively. For cycling, a logarithmic frequency sweep rate shall be used if available; otherwise, other cycling rates of frequency change may be used. The sweep rate shall be such that 15 minutes actual vibration time is required to proceed from the lower to higher and back to the lower extremity of frequency range in accordance with the accelerations or double amplitudes defined in Table VI.

TABLE V

VIBRATION TEST SCHEDULE

Number of Resonances	0	1	2	3	4
Total Vibration Time at Resonance*	-	30 min	1 hr	1-1/2 hr	2 hr
Cycling Time	3 hr	2-1/2 hr	2 hr	1-1/2 hr	1 hr

* 30 minutes each resonance

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TABLE VI
VIBRATION SCHEDULE

Frequency (cps)	5-22	22-40	40-74	74-500
Double Amplitude (in. ± 10%) Minimum	0.2		0.036	
Acceleration (g's ± 10%) Minimum		3		10

4.4 .4.3 Acceleration, ready condition. The igniter, in the ready condition , shall be subjected to the accelerations defined in Table VII after being secured to an acceleration equipment (see Figure 2). Acceleration shall be measured at the approximate geometric center of the igniter.

TABLE VII
ACCELERATION SCHEDULE

Magnitude (g's ± 10%)	Acceleration Direction (See Figure 2)	Time
25	Longitudinal axis, Fwd	1 minute
35	Longitudinal axis, Aft	1 minute

4.4.4.4 Shock, ready condition. The igniter, in the ready condition, shall be attached to a shock test machine (see Figure 2). Shock magnitudes and durations shall be measured by an accelerometer mounted on the mounting fixture. Tests on the igniter shall be performed in accordance with Table VIII in the sequence shown.

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TABLE VIII

SHOCK TEST SCHEDULE

Test Sequence	Peak Magnitude (g's ± 10%)	Duration (ins)	Impact Axis	Direction of Deceleration Force	Number Of Impacts
1	300	0.5 to 1.5	Longitudinal	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	18.0 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

4.4.4.5 Temperature and humidity. non-operating condition. Eight preproduction igniters shall be subjected, in the non-operating condition, to temperature and humidity cycling in the following manner: The specimen shall be placed in a temperature chamber which has been stabilized at an environmental temperature of -45° F +0° -5° F. The specimen shall remain in the environment for 6 hours. (As an alternative procedure, the specimen may be placed in the chamber, with the chamber at room ambient temperature. Chamber temperature shall then be reduced to -45° F +0° -5° F within 2 hours maximum. The specimen shall thereafter remain in the chamber for 4 hours) . The specimen shall then be subjected to an environmental temperature of +140° F +5° -0° F. Maximum transition time from the low temperature to the high temperature shall be 2 hours. The specimen shall remain in the high temperature environment for 4 hours . This cycle shall be performed 3 times; maximum transition time shall be 2 hours. The specimen shall subsequently be subjected to an environment of +110° F +5° -0°F at a minimum relative humidity of 95 percent for a minimum of 8 hours. Thereafter, 4 igniters shall be operated in accordance with 4.4.4.6 and 4 igniters shall be operated in accordance with 4.4.4.7. Prior to operation the igniters may be wiped dry. Drying by an air blast or oven is not permitted.

4.4.4.6 Low temperature operating. Before testing, the igniter, installed in its test chamber in the ready condition, shall be conditioned at a temperature of 0° F ± 5° F for a period of 1 hour. Immediately thereafter the igniter shall be operated in accordance with and meet the requirements of 4.4.4.18.

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4 .4.4.7 High temperature operating. Before testing, the igniter, installed in its test chamber in the ready condition, shall be conditioned at a temperature of +140° F ± 5° F for a period of 1 hour. Immediately thereafter the igniter shall be operated in accordance with and meet the requirements of 4.4.4.20.

4.4.4.8 Auto-ign ition, low temperature. The igniter shall be subjected to a temperature of +170° F ±10° F, relative humidity less than 30 percent, for 2 hours. Combustion shall not occur.

4.4.4.9 Auto-ignition, high temperature. The igniter shall be subjected to a temperature of +250° F ± 20° F, relative humidity less than 30 percent, for 8 hours. Combustion shall not occur. The igniter need not be capable of operation after this test.

4.4.4.10 Vibration, packed and packaged. The igniter, packed and packaged in accordance with Section 5, shall be vibrated in such a manner as to determine the frequency producing maximum transmissibility within the frequency range of 5 to 60 cps. The container shall be fastened, in suitable fashion, to the table of a vibration testing device producing an essentially sine wave motion. The maximum permissible motion freedom shall be approximately 1/32 inch. The specimen shall then be vibrated for 1 hour at the frequency producing maximum transmissibility and at an input acceleration of 1 g. This test shall be applied to each of the three principal orthogonal axes.

4.4.4.11 Shock . Packed and packaged. The igniter, packed and packaged in accordance with Section 5, shall be subjected to free fall drop tests. The igniter in its container shall be dropped from a height of 36 inches. A total of four edgewise drops shall be made. Two drops, each, shall be made on the bottom and top edges at points 180 degrees apart. In addition, flat drops shall be made on an impact surface of essentially non-yielding material such as stone or concrete. The complete series of drops shall be performed under each of the following conditions: room temperature, within 10 minutes after the container has been stabilized at -45° F ± 5° F, and within 10 minutes after the container has been stabilized at +140° F ± 5° F.

4.4.4.12 Jolt. The igniter, in the non-operating condition, shall be jolt-tested in accordance with and shall meet the requirements of MIL-STD-350. The igniter need not be capable of operation after the jolt test.

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4.4.4.13 Jumble. The igniter, in the non-operating condition, shall be jumble-tested in accordance with and shall meeet the requirements of MIL-STD-351. The igniter need not be capable of operation after the jumble test.

4.4.4.14 Drop. The igniter, in the non-operating condition, shall be drop-tested in accordance with and shall meet the requirements of MIL-STD-352. The igniter need not be capable of operation after the drop test.

4.4.4.15 Transportation vibration. The igniter, in the non-operating condition, shall be vibrated in accordance with and shall meet the requirements of MIL-STD-353.

4.4.4.16 Accelerated storage life, packed and packaged. condition. The igniter, in the packed and packaged condition, shall be subjected to an environmental temperature of +140° F +5° -0° F, at ambient atmospheric conditions, for a period of 30 days.

4.4.4.17 Accelerated storage life, ready condition. The igniter, in the ready condition, shall be subjected to an environmental temperature of +140° F +5° -0° F, at ambient atmospheric conditions for a period of 14 days.

4.4.4.18 Minimum firing current. A direct current of 0.9 ± 0.01 amperes shall be passed for 100 ± 1.0 ms through the bridge wire between pins 1 and 2 or the bridge wire between pins 3 and 2. The igniter shall fire and shall exhibit the characteristics of 4.5.6.2 b.

4.4.4.19 Maximum no-fire current. A direct current of 0.2 ± 0.02 ampers shall be passed through either bridge wire for a minimum of 5 minutes. The igniter shall not fire.

4.4.4.20 Maximum firing current. A direct current of 5 ± 0.2 amperes shall be passed through the bridge wire between pins 1 and 2 or through the bridge wire between pins 3 and 2 for 20 ± 1.0 ms. The igniter shall fire and shall exhibit the characteristics of 4.5.6.2 b.

4.4.5 Test data sheets. All test data shall be recorded on a data sheet by the supplier and maintained as part of the inspection records. In addition to the data requirements of 4.5.7 the data sheet shall contain, but are not limited to, the following requirements:

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a. Minimum firing current _____
b. Maximum no-fire current _____
c. Maximum firing current _____

4.4.6 Determination of defect or failure. Failure of an igniter to successfully pass any of the tests in 4.4 through 4.4.4.20 shall be cause for rejection of the preproduction sample.

4.5 Acceptance inspection. Igniters produced under the contract or purchase order shall be subjected to the following tests and examinations.

4.5.1 Tests and examinations.

4.5.1.1 Prior to firing.

a. Certification of data
b. Static pressure test
c. Insulation resistance
d. Bridge wire resistance
e. Configuration

4.5.1.2 Firing.

a. Firing current, two elements
b. Pressure output

4.5.1.3 Post-firing.

a. Static pressure test
b. Insulation resistance
c. Pin-to-pin resistance

4.5.2 Each igniter delivered under the contract shall be subjected to the tests in 4.5.6,1 a. through d.

4.5.3 A sample shall be selected using the plans and procedures set forth in MIL-STD-105, Inspection Level II and subjected to the additional tests and examinations in 4.5.6.1 e., 4.5.6.2 and 4.5.6.3. An AQL of 1.0 shall be used for 4.5.6.1 e. except where otherwise Indicated. An AQL of 0.1 shall be used for 4.5.1.2 and 4.5.1.3.

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4.5.4 Test conditions. Unless otherwise specified, the igniter shall be subjected to the acceptance tests under the following conditions:

a.	Temperature:	Room ambient -32° F to +98° F
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

4.5.5 Test equipment and set-up. The test equipment set-up shall be in accordance with Figure 3.

a. Power supply: A constant current power supply capable of supplying 10 ± 0.1 ampere, 20 ± 1 ms pulse, through a load of 0.6 ohm
b. Oscilloscope: 5-inch (and camera)
c. Pressure transducer: Pressure range 0 to 5000 psig, accuracy to ± 1.0 percent full scale
d. Resistance measuring devices:

(1) Ohmmeter, range 0-2000 ohms, accuracy ± 1/2 percent full scale, maximum current 50 ma, maximum voltage 1.5 volts, direct current.
(2) Resistance bridge, accuracy ± 1/2 percent, readout resolution 0.1 ohms minimum, maximum current 13 ma.

e. Hydraulic test stand with gage: 0 to 10,200 psig minimum

4.5.6 Test procedures.

4.5.6.1 Prior to firing.

a. Certification of data.

(1) Calorimetric output, The supplier shall certify that a calorimetric bomb test has been performed on each powder lot and that the calculated calorimetric output of each explosive charge assembled into a igniter is not less than 900 calories.
(2) Metallic particles. The supplier shall certify that the main charge contains a minimum of 25 percent, by weight, of metallic particles.
(3) Ammunition data cards. Ammunition data cards shall be supplied with each lot of igniters.

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b. Static pressure test . The igniter case with the connector pins bonded in place shall be pressure-tested prior to assembly. A hydraulic fluid shall be applied to the inside of the case at a pressure of 10,000 ± 100 psig for a minimum of 10 seconds. There shall be no leakage from either the case or the bond at the pin connections during the test. There shall be no distortion of the case as a result of this test. After completing the test, the case and connector shall be thoroughly cleaned and dried.

c. Insulation resistance (prior to firing). The resistance between each pin contact (1, 2 and 3) and the case shall be 1000 ohms minimum at 1.5 volts, direct current maximum.

d. Bridge wire resistance (Prior to firing). The resistance between pins 1 and 2 shall be 1.2 ± 0.15 ohms. The resistance between pin 2 and pin 3 shall be 1.2 ± 0.15 ohms .

e. Configuration.

Dimensions. Refer to BuWeps Dug. 2526678.

Threads:

9/16-18 UNF-3 major dia MIN
9/16-18 UNF-3A pitch dia

Connector:

Verify that connector pins conform to the dimension requirements of MS 3192, Notes 1, 3 and 5 of the drawing.

Other:

.083+ .015 - .000 inches
.481 + .002 - .003 dia inches
.600 ± .020 inches

Materials:

Case and connector pin material shall be steel, AISI B1113; Alternate: QQ-S-633

Visual:

Packing, preformed	Shall be free from defects, Ref. BuWeps Dug. 2526718-2
Chamfer (threads)	Verify presence of 30" chamfer
Plating	Verify cadmium plating of case and connector pins in accordance with QQ-P-416, Type II, Class 2

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Identification: The identification markings shall be correct, legible and in accordance with the drawing. AQL 2.5.

4.5.6.2 **Firing.**

a. Firing current - two elements. With the bridge wires connected in parallel (pins 1 and 3 connected together) a direct current, totaling 10 ± 0.5 ampers, shall be passed through the bridge wires for 20 ± 1 milliseconds and shall cause the igniter to fire.

b. Pressure output. With the igniter and test chamber at atmospheric pressure and at a temperature within the range of +0° F ± 5° F to +140° F ± 5° F, and upon application of firing current, the igniter shall develop a pressure output versus time characteristic within the limits illustrated in Figure 1 when operated into a closed chamber of 0.305 cubic inches.

4.5.6.3 Post firing.

a. Static pressure test. A hydraulic pressure shall be applied to the end of the case at a pressure of 10,000 ± 100 psig for a minimum of 10 seconds. There shall be no detectable fluid leakage from either the case or the bond at the pin connections during the test. There shall be no distortion of the case as a result of this test.

b. Insulation resistance. The resistance between each pin contact (1, 2 and 3) and the case shall be 1.05 ohms minimum at 1.5 volts, direct current maximum.

c. Pin-to-pin resistance. The resistance between pin 1 and pin 2 shall be 1.05 ohms minimum. The resistance between pin 2 and pin 3 shall be 1.05 ohms minimum at 1.5 volts, direct current maximum.

4.5.7 Test data sheets. All test data shall be recorded on data sheets by the supplier. Data sheets and photographic records shall be maintained as part of the inspection records. Data sheets shall contain, but are not limited to, the following information:

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Date _____

Prior to firing

a. Certification of data _____
b. Static pressure _____ psig
c. Insulation resistance _____ ohms
d. Bridge wire resistance _____ ohms
e. Configuration _____

Firing

a. Firing current _____ amps
b. Insulation resistance _____ ohms

Post firing

a. Static pressure _____ psig
b. Insulation resistance _____ ohms
c. Pin-to-pin resistance _____ ohms

4.5.8 Determination of defect or failure. A defective is a unit of product which contains one or more defects. Failure of an igniter to successfully pass any of the tests in 4.5.6.1 e. through 4.5.6.3 c. shall constitute a defect. Failure of an igniter-to pass any of the-tests in 4.5.6.1 a. through d. shall be cause for rejection of that igniter.

4.6 Periodic production inspection. Periodic production samples shall be selected in accordance with 4.3.3. The samples shall be tested in accordance with the following plan:

a. The Sample of 9 units from the first group of 150 units will be called the first Sample and will undergo the tests *of* Table IX for Sample 1. The sequence of tests will be as shown in Table IX.

b. The Sample of 9 units from the second and third groups of 150 units will be called the second and third Samples and will undergo the tests of Table IX for Samples 2 and 3 respectively. The sequence of tests will be as shown in Table IX.

c. The next Sample will be selected from the next group of units regardless of whether the shift to Step 2 was made. This sample will be called the fourth Sample and will undergo the tests of Table IX for Sample 4, etc.

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TABLE IX

PERIODIC TEST SEQUENCE

Tests	Sample and Test Sequence 1, 4, Etc. (3)	(3)	(3)	2, 5, Etc. (3)	(3)	(3)	3, 6, Etc. (3)	(3)	(3)
Inspection, Prior to Firing	1	1	1	1	1	1	1	1	1
Vibration, Ready Condition					2*				2*
Acceleration, Ready Condition									3*
Shock, Ready Condition								2*	4*
Temperature and Humidity		2						3	5
Jolt							2		
Jumble				2					
Forty Foot Drop	2								
Transportation Vibration									
Low Temperature Operating								4	
High Temperature Operating						2			
Auto-Ignition, Low Temperature		3							6
Auto-Ignition, High Temperature			3				3		
Accelerated Storage Life, Ready Condition					3				
Maximum No-Fire Current					4				7
Minimum Firing Current					5				
Maximum Firing Current		4							
Inspection, Post Firing		5			6	3		5	9

* Visual Check After Testing

4.6.1 Test conditions. Unless otherwise specified, the igniter shall be subjected to the periodic tests under the following conditions:

a.	Temperature:	Room ambient +32° F to +98" F
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

4.6.2 Test equipment and set-up. The test equipment and set-up for the periodic tests shall be in accordance with 4.4.3.

4.6.3 Test procedures.

4.6.3.1 Inspection, prior to firing. Inspection of the igniter, prior to firing, shall be performed in accordance with 4.5.6.1.

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4.6.3.2 Vibration , ready condition. Vibration testing of the igniter, in the ready condition, shall be performed in accordance with 4.4.4.2.

4.6.3.3 Acceleration. ready condition. Acceleration testing of the igniter, in the ready condition, shall be performed in accordance with 4.4.4.3.

4.6.3.4 Shock, ready condition. Shock testing of the igniter, in the ready condition, shall be performed in accordance with 4.4.4.4. ,

4.6.3.5 Temperature and humidity, non-operating condition. Temperature and humidity testing of the igniter, in the non-operating condition, shall be performed in accordance with 4.4.4.5.

4.6.3.6 Jolt . Jolt testing of the igniter, in the non-operating condition, shall be performed in accordance with 4.4.4.12.

4.6.3.7 Jumble . Jumble testing of the igniter, in the non-operating condition, shall be performed in accordance with 4.4.4.13.

4.6.3.8 Forty foot drop. The igniter, in the non-operating condition, shall be subjected to the forty foot drop test in accordance with 4.4.4.14.

4.6.3.9 Transportation vibration. The igniter, in the non-operating condition, shall be subjected to the transportation vibration test in accordance with 4.4.4.15.

4.6.3.10 Low temperature operating. The igniter shall be subjected to the low temperature operating test in accordance with 4.4.4.6.

4.6.3.11 High temperature operating. The igniter shall be subjected to the high temperature operating test in accordance with 4.4.4.7.

4.6.3.12 Auto-ignition, low temperature. The igniter shall be subjected to the auto-ignition, low temperature test in accordance with 4.4.4.8.

4.6.3.13 Auto-ignition , high temperature. The igniter shall be subjected to the auto-ignition, high temperature test in accordance with 4.4.4.9.

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4.6.3.14 Accelerated storage life, ready condition. The igniter shall be subjected to the accelerated storage life test, in the ready condition, in accordance with 4.4.4.17.

4.6.3.15 Maximum no-fire current. The igniter shall be subjected to the maximum no-fire current test in accordance with 4.4.4.19.

4.6.3.16 Minimum firing current. The igniter shall be subjected to the minimum firing current test in accordance with 4.4.4.18.

4.6.3.17 Maximum firing current. The igniter shall be subjected to the maximum firing current test in accordance with 4.4.4.20.

4.6.3.18 Inspection, post firing. The igniter shall be subjected to the post firing inspection in accordance with 4.5.6.3.

4.6.4 Test data. Test data shall be recorded in accordance with 4.4.5, as appropriate, and maintained as a part of the inspection record.

4.6.5 Determination of defect or failure. Failure of the igniter to successfully pass the tests in 4.6.3.1 through 4.6.3.18 shall be cause for withholding acceptance of all units on hand. Corrective action shall be taken in accordance with 4.3.3.

5. PREPARATION FOR DELIVERY

5.1 The igniter shall be packed and packaged in accordance with instructions listed on BuWeps DL 1330411.

5.2 Marking.

5.2.1 Special marking. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

5.2.2 Normal marking. In addition to the special markings and the markings required by the contract or purchase order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

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b. NOTES

6.1 Intended use. The igniter is intended for use in a torpedo propulsion system to initiate the starting sequence of a positive displacement engine.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. Assigned activity for preproduction tests
c. Any exceptions to this specification
d. Inspection data required
e. Ammunition data card requirement and distribution of data cards
f. Hazard classification

6.3 Definitions .

6.3.1 Ready. The condition of the igniter when fully assembled in its housing or equivalent, prior to application firing current.

6.3.2 Operating. The condition of the igniter, fully assembled in its housing or equivalent, from the instant of application of firing current to the termination of generation of igniter combustion products.

6.3.3 Non-operating. The condition of the igniter when stored, transported, or handled. Tests on the igniter, in the non-operating condition, shall be performed without packaging or packing.

6.3.4 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak or the corresponding maximum acceleration in gravitational units. The frequency is given in cycles per second.

6.3.5 Acceleration (steady state). Acceleration (steady state) is the time race of change of velocity of a body resulting from the application of a sustained external force with a uniform build-up to peak in more than 0.010 seconds. Acceleration is given in gravitational units.

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6.3.6 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock, in general, can be described as a build-up to 90 percent of peak force in less than 0.010 seconds. Shock is described as a pulse in units of gravitational acceleration, duration in milli-seconds and shock waveform equivalent to one half sine wave. Duration is defined as the interval between the point of zero amplitude, from which the shock waveform builds up, to its decay to zero amplitude.

6.3.7 Temperature. The temperature of the environment to which the igniter is exposed, is expressed in degrees Fahrenheit.

6.3.8 Humidity. Humidity is the relative humidity of the environ-ment to which the igniter is exposed, expressed in percent.

6.3.9 Metallic particles . Metal oxidants used in the main charge.

6.3.10 Main charge. All pyrotechnic material in the igniter exclusive of the primer material.

6.4 Weight. The weight of the igniter shall not exceed 2 ounces.

6.5 General safety precautions.

6.5.1 The loading, assembly, and handling of the explosive components of the igniter covered by this specification involve hazardous operations and therfore require the observance of suitable safety precautions. Use of this specification shall not be construed to relieve the contactor of the manufacturer of responsibility for the safety of his operations. Listed below are the minimum provisions which a contractor or manufacturer of the propellant should observe in order to fulfill his responsibility for safety. At Bureau of Naval Weapons, Navy Department, and other Government plants these provisions are mandatory.

6.5.1.1 All handling operations should be conducted in a neat and orderly manner.

6.5.1.2 Personnel handling the propellant should, insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion or deterioration from perspiration or other contaminating deposits.

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6.5.1.3 The exposure of propellant materials should be controlled to minimize the absorption of moisture from the atmosphere or other sources during handling operations.

6.5.1.4 All explosives should be stored in suitable storage magazines located in accordance with the American Table of Distance (A.T.D.). All storage should be in adequate ready of service magazines, located in accordance with interplant distances.

6.5.1.5 Safe equipment and methods should be utilized for transporting and handling explosives.

6.5.1.6 Proper care should be exercised at all times to protect personnel from accidents, fires, or explosions and to limit damage to equipment. In this connection, the following precautionary measures should be observed.

a. Keep only minimum quantities of explosives present at each stage of operation.
b. Keep propellant and explosive parts of the gas generator in approved, covered receptacles with covers in place when material is not being taken out or put into the receptacles. Where necessary, receptacles should be conductive in order to ground electrostatic charges.
c. Protect all operations from electrostatic charges by effectively grounding all machinery, equipment and fixtures and where necessary, provide suitable grounded conductive coverings on floors, work benches, tables and worker' s shoes.
d. Enforce, where necessary, the wearing of suitable footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning, and similar hazards.
e. Allow no fires or exposed sparking of electrical equipment and little or no flammable material to be present in handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.
f. Enforce good housekeeping and policing methods throughout the handling area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosive or flammable material, and assure its removal from floors, walls, ceilings, ledges, tables, benches and work equipment. Clean up any spilled material immediately.

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g. Employ properly proportioned and properly located protective barricades, screens, or shields at all required points.

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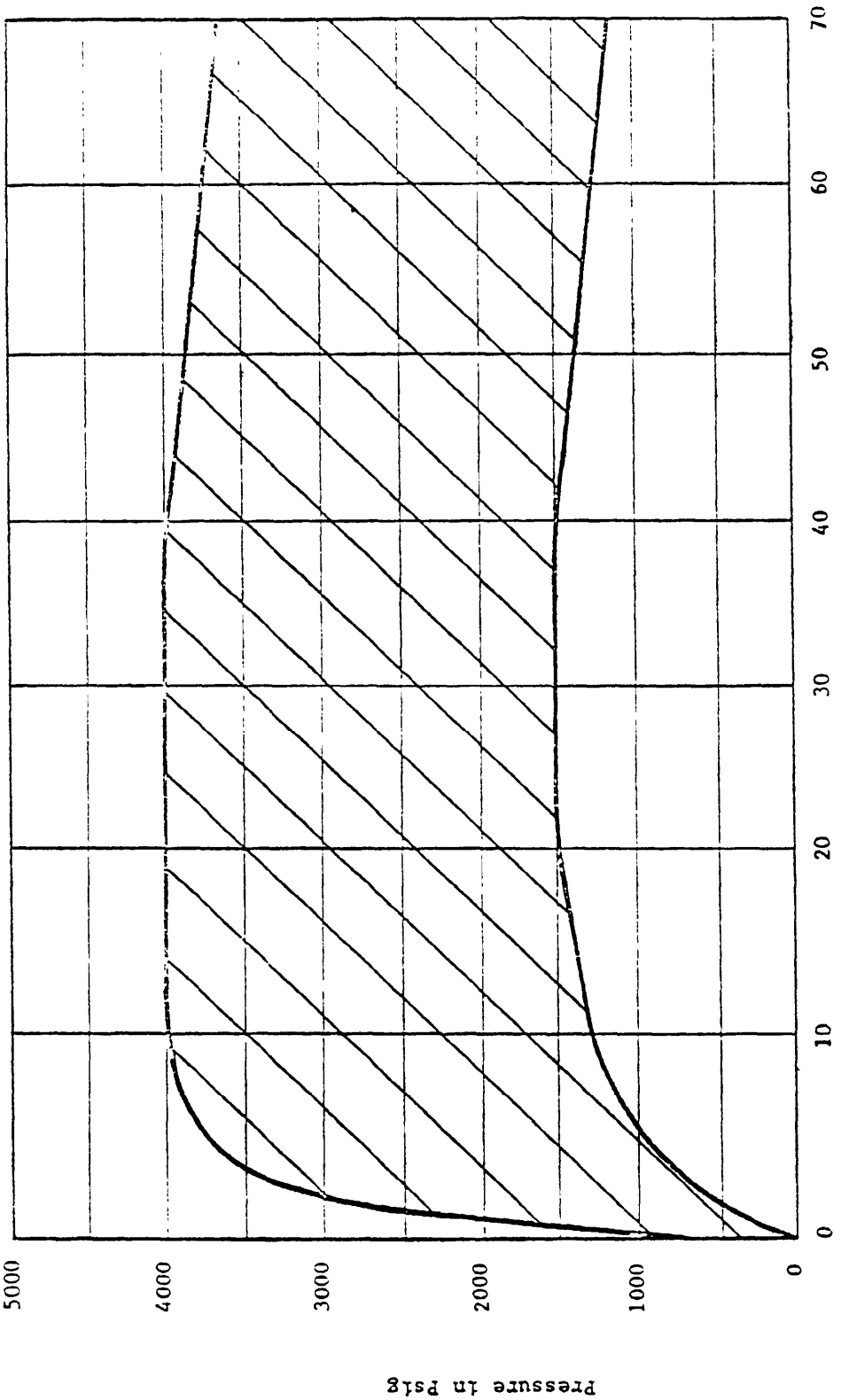


Figure 1, Pressure-Time Envelope

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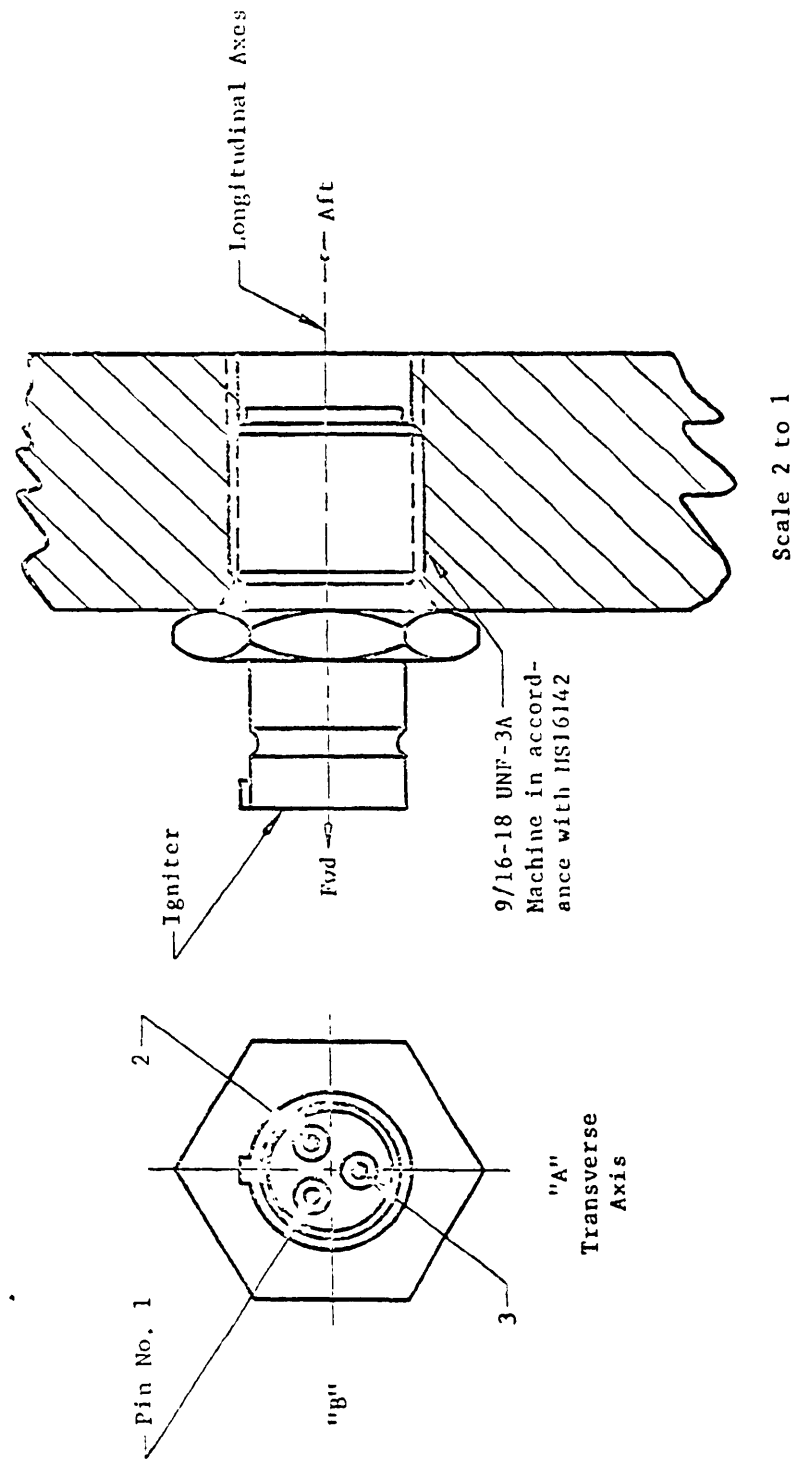


Figure 2, Igniter Mounting

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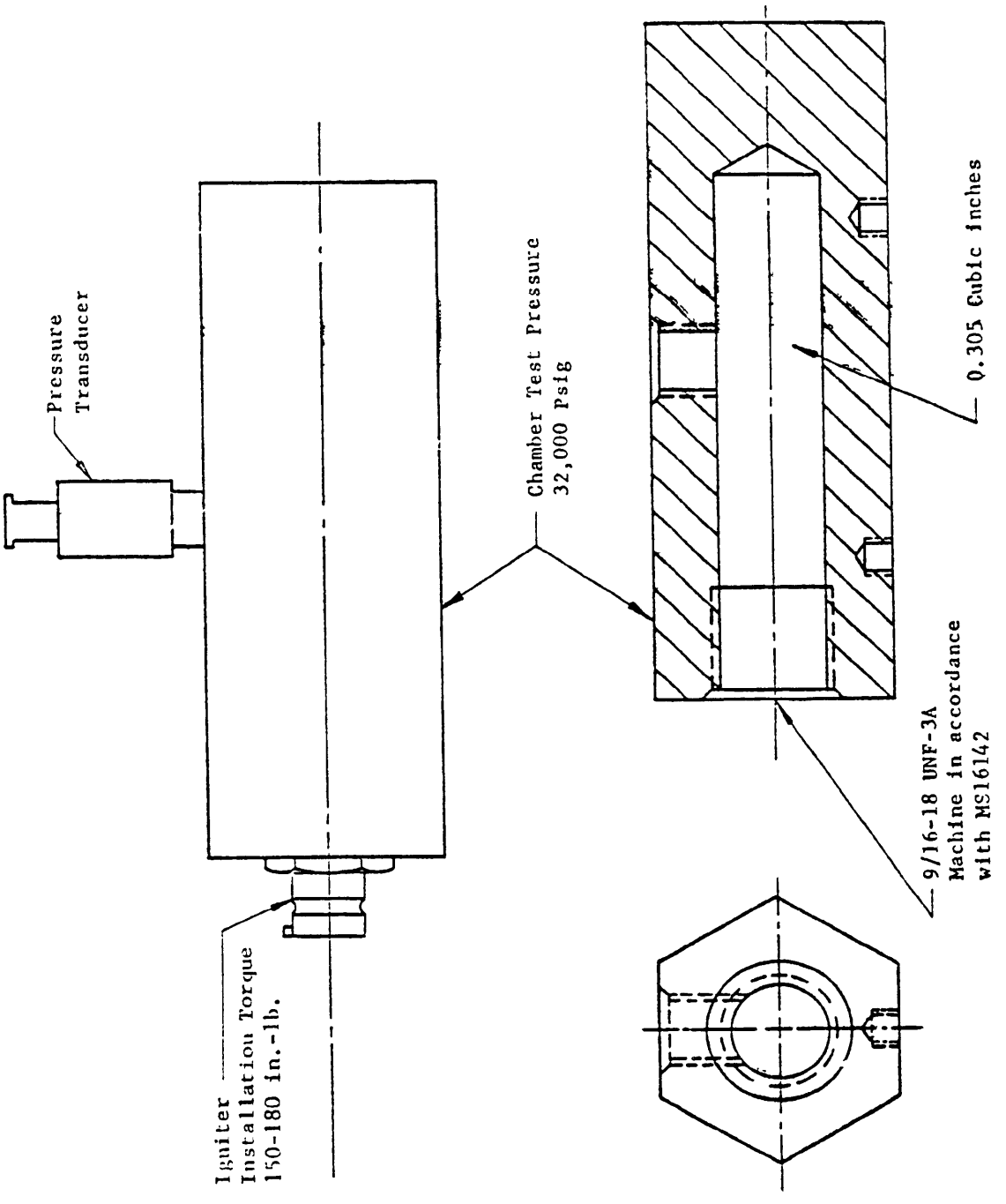


Figure 3, Pressure Test Chamber

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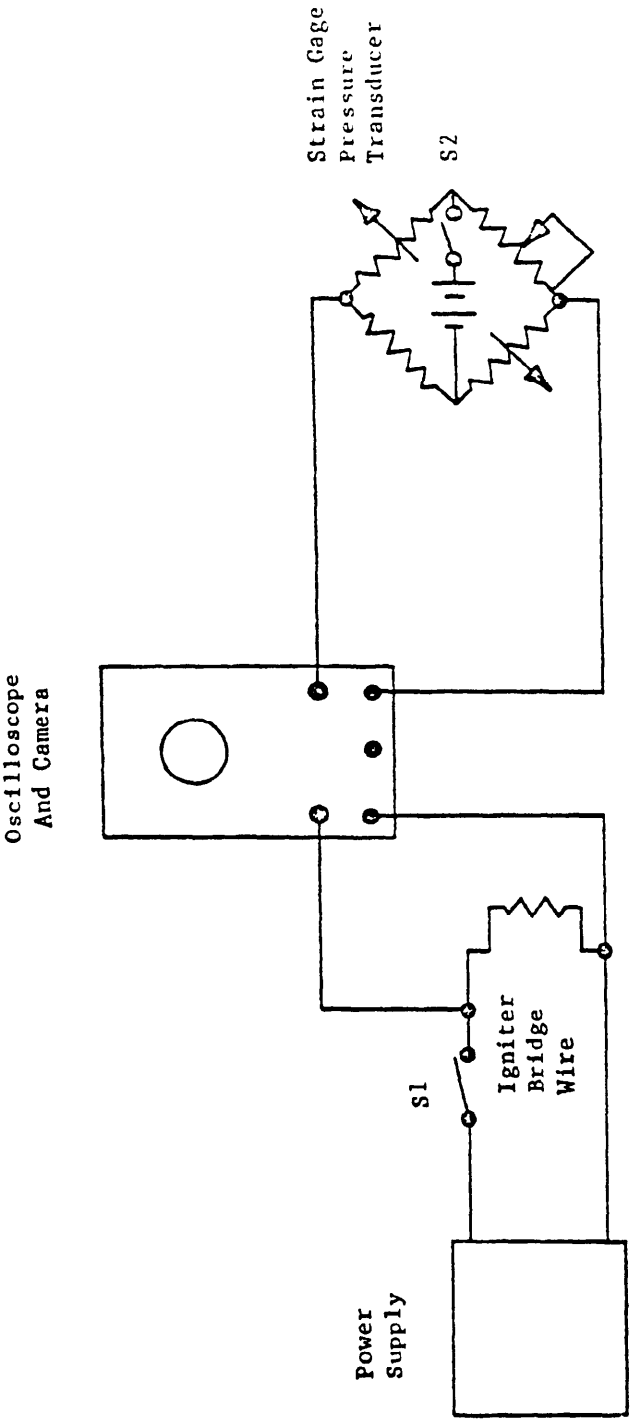


Figure 4, Igniter Test Schematic

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SPECIFICATION ANALYSIS SHEET

Form Aporoved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

MIL-I-82437(OS) IGNITER

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1 APR 63 1426

C-8270

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Naval Ordnance Systems Command
Washington, D. C. 20360

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MIL-I-82437(OS)
Amendment 3
29 April 1969
Superseding
Amendment 2
14 August 1968

MILITARY SPECIFICATION

IGNITER

This amendment forms a part of Military Specification MIL-I-82437(OS) of 10 May 1966 and has been approved by the Naval Ordnance System Command, Department of the Navy

Wherever it appears in specification delete “Bureau of Naval Weapons” and substitute “Naval Ordnance Systems Command”: also delete the term “Acceptance Tests” and substitute “Quality Conformance Tests”.

Page 1, paragraph 2.1, under SPECIFICATIONS: Delete "MIL-A-9836" and its associated title; delete "QQ-S-633" and associated title and substitute "QQ-S-637 Steel, Bar, Carbon, Cold Finished (Standard Quality, Free Machining)”; under STANDARDS, add "MIL-STD-1167 Ammunition Data Card”.

Page 3, add the following new paragraph:
“3.2.5 Safety. The supplier or manufacturer, as may apply, shall provide adequate safety precaution for protection of personnel during the handling of igniters (see 6.5)”.

Page 8, paragraph 3.6: Delete "MIL-A-9836" and substitute "MIL-STD-1167".

Page 9, paragraph 4.3.3d. : Delete "activity" and substitute “activity”.

Page 14, paragraph 4.4.4.5, first sentence. Delete “Eight preproduction igniters shall be subjected” and substitute “The igniter shall be subjected”; in lines 17, 18, and 19 delete the following sentence: “Thereafter, 4 igniters shall be operated in accordance with 4.4.4.6 and 4 igniters shall be operated in accordance with 4.4.4.7."

Page 19, paragraph 4.5.6.1e. , under subheading "Materials:": Delete "QQ-S-633" and substitute "QQ-S-637".

Page 20, paragraph 4.5.6.2a., line 3: Delete "ampers" and substitute “amperes”.

Page 20, paragraph 4.5.7, line 2: Delete “suppler” and substitute "supplier" .

FSC 1355

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Page 21, paragraph 4.6a. , last sentence: Delete “Table IX. ” and substitute “Table IX. (Col II)”.

Page 21, paragraph 4.6b., last sentence: Delete “Table Ix.” and substitute “Table IX (Cols III and IV)”.

Page 21, paragraph 4.6c last sentence: Delete “Sample 4, etc.” and substitute “Sample 4, etc. (Col II)”.

Page 22. Replace Table IX with new Table IX attached.

pages 26 through 28. Delete paragraph 6.5 and its subparagraphs and substitute the following:

“6.5 Explosives safety precautions

6.5.1 Applicability. The loading, assembly, and handling of the explosives, subassemblies, and the finished items covered by this specification, involve hazardous operations and therefore require suitable explosives safety precautions. At plants of the Naval Ordnance System Command, Department of the Navy, and other Government plants where the government is responsible for safety, the precautions listed below are mandatory. These same minimum provisions listed below are recommended for observance by a contractor or manufacturer that explosively loads the item covered in this specification, in order to fulfill his own safety responsibility. Other warnings and precautions pertaining to the operational effectiveness or safety of the specified item during its use (for example, during its handling, launching, or firing) are included in the detailed technical requirements of the specification. Use of this specification shall not be construed as replacing other safety regulations of Governmental agencies or as relieving the contractor or manufacturer of responsibility for the safety of his operations.

6.5.2 Personnel and equipment. Proper care must be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In connection with the above, the precautionary measures in-the following paragraphs should be observed. (See also 6.5.4, Static and electrical, and other applicable parts of these safety precautions.)

6.5.2.1 Keep explosives operations under strict supervision. Periodic inspections shall be employed to ensure compliance with established safety regulations.

6.5.2.2 Use safe equipment and methods when handling and transporting explosives and loaded parts.

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6.5.2.3 Where required, use remote control, barricaded handling equipment for explosives operations such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.5.2.4 Employ properly proportioned and properly located protective barricades, screens, or shields at all required points.

6.5.2.5 Maintain equipment such as presses, dies, crimpers, blenders, mixers, sifters, conveyors, and the like, in safe operating condition.

6.5.2.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against bums, poisoning, and associated industrial hazards.

6.5.2.7 During static tests, restrain propulsion units such as rocket motors against full thrust to prevent flight if ignited (for example, during test of igniter circuit). Also employ means for harmlessly venting flames and exhaust gases.

6.5.3 Operating procedures. All loading operations shall be conducted in full accordance with a formalized operating procedure that shall include the following:

6.5.3.1 Keep only minimum, limited quantities of explosives and completed or partially loaded parts present at each stage of operations.

6.5.3.2 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling, and storage spaces. Enforce proper “Match” and "No Smoking” rules where necessary.

6.5.3.3 post appropriate safety instructions for specific operations and instruct personnel as to the hazards present and precautions involved.

6.5.3.4 Establish and post in each work space the number of working personnel and transients permitted in such areas. The number established shall be the minimum number required to accomplish necessary operations.

6.5.3.5 Conduct all loading operations in a neat and orderly manner.

6.5.3.6 Enforce good housekeeping procedures and maintain effective policing, inspection, and supervisory methods throughout the loading area and surroundings.

6.5.3.7 Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon,

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and to ensure its removal from, floors , walls, ceilings , ledges, tables, benches, piping, and equipment or the items loaded. Clean up any spilled material immediately.

6.5.3.8 Control the exposure of explosive materials and related parts so as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.3.9 Ensure that personnel handling detonators, primers, delay elements, lead-ins, booster, and related parts which affect functioning, avoid, insofar as practicable, using bare fingers or improper equipment, in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.5.3.10 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges. (See 6.5.4)

6.5.4 Static and electrical

6.5.4.1 protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches, tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges shall be employed. Fabrics, such as silk and nylon, which promote static generation shall be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low energy type electric primers, detonators, and squibs. The latter types of items shall have the free ends of lead wires bared and twisted together, and shall be packed in relatively small groups wrapped in bare, noninsulated aluminum foil or other uncoated metal foil. During assembly and processing operations, such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding from induced electric currents originated by sources such as lightning, static, electromagnetic radiations, etc. Where necessary for safety, humidity in work rooms should be appropriately increased, as required, to lessen electrostatic effects but without inducing excessive moisture absorption by any of the components of the item being processed.

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6.5.4.2 Protect all explosive operations from effects of electric currents originating in equipment such as soldering irons, or guns, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding, separation, or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to the completion of an electric circuit through an electric primer, by. detonator or squib. For example, circuits may be inadvertently completed from a defective electric soldering iron through a grounded contact. All electric type primers, detonators, or squibs provided with wire leads shall have the free ends of the wire bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Removable short circuiting clips, or other devices should be employed during manufacturing operations involving items such as electric primers, detonators, or squibs.

6.5.4.3 Where necessary, protect personnel and operations from electro-magnetic energy. In certain work with very sensitive explosives or items, a shielded environment free from significant electromagnetic energy should be provided.

6.5.5 Operating facility and storage regulations. All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance, at a minimum, with the American Table of Distances (ATD) and, while in process, in safety lockers and chests if in loading rooms or in adequate ready or service magazines located in accordance with Intraplant Distances set forth in the ATD when outside of loading rooms. For Navy-managed or contractor-managed Navy-owned explosives loading plants, the Armed Services Explosives Safety Board standards approved by the Department of Defense, the Secretary of the Navy, and the Naval Ordnance Systems Command, covering quantity-distance relations for explosives, apply."

Page 29. Replace Figure 1 with new Figure 1, attached.

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Table IX

PERIODIC TEST SEQUENCE

I Tests	Sample and Test Sequence II 1, 4, Etc.			III 2, 5, Etc.			IV 3, 6, Etc.		
	(3)	(3)	(3)	(3)	(3)	(3)	(3)	(3)	(3)
Inspection, Prior to Firing	1	1	1	1	1	1	1	1	1
Vibration, Ready Condition					2*				2*
Acceleration, Ready Condition									3*
Shock, Ready Condition								2*	4*
Temperature and Humidity		2						3	5
Jolt							2		
Jumble				2					
Forty Foot Drop	2								
Transportation Vibration			2						
Auto-Ignition, Low Temperature		3							6
Auto-Ignition, High Temperature			3				3		
Low Temperature, Operating		4						4	
High Temperature, Operating						2			
Accelerated Storage Life, Ready Condition					3				
Maximum No-Fire Current					4				7
Minimum Firing Current					5				
Maximum Firing Current									8
Inspection, Post Firing		5			6	3		5	9

*Visual Check After Testing

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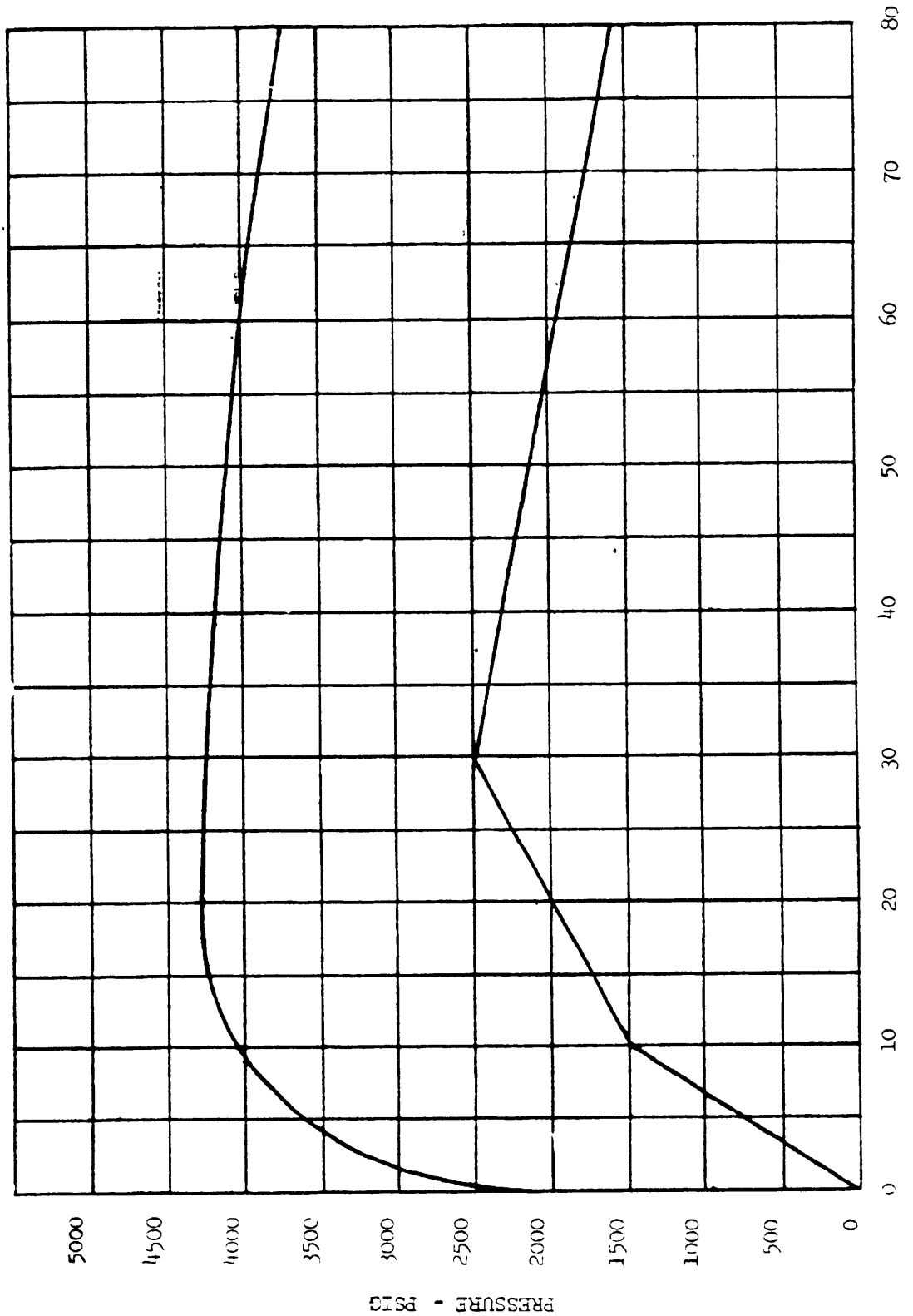


FIGURE 1. PRESSURE-TYPE ENVELOPE

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